

The Foundations of the Mentalist Theory and the Statistical Machine Learning Challenge: Comments on Matthias Mahlmann's *Mind and Rights*

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1. A Human Peculiarity

The contemporary idea of human rights is, for many, intuitive and ordinary; an expression of familiar personal convictions. Seen through the right intellectual lens, however, human rights are a *strange* phenomenon without analog in the animal world. Recognising that this cliché idea is a peculiarly *human* creation allows its underlying moral complexity to come to light. One of the central accomplishments of Matthias Mahlmann's *Mind and Rights: The History, Ethics, Law and Psychology of Human Rights* (*M&R*) is turning this familiar idea into a problem to be solved—a wide-ranging and productive effort “to fathom the nature of morality.”¹

A persistent theme in *M&R* concerns the sophistication of an analytical theory of morality. That is, the degree to which a theory of morality takes the complex nature of moral judgment and its relationship with human rights beliefs seriously. To this end, Mahlmann blends a historical analysis of moral judgments and their instantiation in the law, a philosophical inquiry into the justification of a theory of human rights, and—the theme with which we are most interested here—the foundation of the moral judgments giving rise to human rights in the human mind, articulated within the “mentalist” theory of law and ethics.

I am broadly sympathetic to Mahlmann's effort to link the idea and history of human rights with the mentalist theory.² Specifically, I maintain that the Universal Moral Grammar (UMG) research program is the proper lens through which to study moral cognition and is rightfully brought into interaction with a theory of human rights.

That said, this discussion of *M&R* is occurring when the research program from which UMG takes its methodological cues—Universal Grammar (UG) in linguistics—is undergoing renewed scrutiny regarding the implications of *statistical machine learning* for language acquisition. The statistical learning challenge to UG is not by any means novel. Nor is the machine learning challenge. What *is* new is the rise of transformer-based Large Language Models (LLMs) and their

¹ Matthias Mahlmann, *Mind and Rights, The History, Ethics, Law and Psychology of Human Rights* (Cambridge University Press 2023) 418.

² Vincent J. Carchidi, ‘The Nature of Morals: How Universal Moral Grammar Provides the Conceptual Basis for the Universal Declaration of Human Rights’ (2020) 21 Human Rights Review 65-92.

human-like language outputs and novel efforts to simulate language acquisition via developmentally realistic means.

Given that the reasoning in support of UMG takes inspiration from UG, a fundamental challenge to the latter could plausibly be transformed into a fundamental challenge to the former, too, and thus the mentalist theory of law and ethics.³ For this reason, I take the statistical machine learning challenge to the mentalist (or generative) approach as the central topic of my contribution to this symposium on *M&R*.

There are deeper reasons for doing this: computational modelling of human cognitive capacities is at risk of contributing to those theses identified in *M&R* as ‘far-reaching revisionist theories, holding that the arguments about the justification of human rights that have been mustered here are in fact no more than an intellectual hoax...’⁴ Conspicuously, the debate that has sprung up in linguistics over such modelling is littered with conceptual confusions that have persisted for years in the case of UG. One can imagine similar confusions arising vis-à-vis UMG.

I thus aim to make clear the theoretical foundations of UMG and UG through this challenge. The driving question of this chapter is: *What are the scope and limits of computational modelling in fundamental theoretical debates in moral psychology and linguistics?*

I begin with an outline of *M&R*’s central argument. Then I turn to the theoretical foundations of UMG, explicating the central arguments and clarifying the kind of concept entailed by UMG through an explication of UG. Here, I devote attention to the dispute over whether language in the mind is *domain-general* or *domain-specific*. From there, the paper dives into the challenges from LLMs and the small-data language modelling challenge. The central arguments tie into a core theme in *M&R* related to the necessity of a rich *descriptive* account of morality and the *theory-dependence of data-interpretation*. More specifically, arguments in response to computational modelling revolve around the *format* of the input, the *sequence* of learning trajectories, the *constraints* on learning, the *means* by which data is *collected*, and the *nature of the use* of acquired linguistic knowledge. The *scaffolding* of models’ learning by human engineers such that basic cognitive development is circumvented emerges as a critical point.

2. Mind and Rights

The *concept* of a right begins Mahlmann’s effort, arguing that rights ‘are nothing less than a fundamental element of human beings’ moral world.’⁵ *Human* rights, specifically, are created by human beings on the basis of normative incidents, like claims made on others. Such claims ‘empower the rights-holders’⁶ by enabling them to ‘justifiably demand something of others.’⁷ In this view, human rights are not mere privileges, as they always involve *claims* made by holders on

³ Ultimately, all we have at our disposal are *arguments*, better and worse. I aim to offer some of the better ones here.

⁴ Mahlmann, *Mind and Rights* (n 1) 325.

⁵ Ibid 45.

⁶ Ibid.

⁷ Ibid 46.

addressees and the latter's corresponding *duties*. Human rights thus consist of a 'cluster of four normative positions: the claims and privileges of the bearer and the duties and no-rights of the addressee.'⁸

Human rights are a product of self-reflectively induced burdens that secure for all humans—by virtue of their humanity—an 'equally shared opportunity to lead an autonomous human life.'⁹ The normative structure of human rights, Mahlmann argues, can be historically identified across cultures. The case of an illiterate Herero woman who expresses that her mistreatment and rape by German soldiers is wrong *without* the language of rights nonetheless has the idea, in whatever obscure form, that what is happening to her is not right, that she should be free to be left alone, to not have her body and inner self harmed by such acts, that she, as a human being, can rightly demand not to be treated like this, neither by state forces nor by private parties, and that everybody is obliged to abstain from hurting her in these pernicious ways.¹⁰

This possession of specific normative content echoes a continuous theme in Mahlmann's historical overview, in which individuals make a claim to be treated a certain way that correlates with a duty by others to act consistently with this claim; '*a specific set of correlated normative incidents*'¹¹ which reflect an 'intricate normative world'¹² that Mahlmann holds to be convincingly evident in indigenous and non-native societies alike.

Mahlmann is careful to separate moral *intuitions* from the contemporary legal notion of human rights, illustrating a process leading from basic moral judgments to the contemporary human rights landscape. It begins with *principled moral intuitions* that have claim-related content, followed by a perception that others have a duty that *corresponds* with this claim. The individual experiencing these first two steps understands themselves, furthermore, as being able to decide whether to *act* materially within the scope of this claim. Moving from this understanding is the *generalisation* of the abstract ethical content related to said claims and, then, the *universalisation* to all humans of the claim. *Rights* eventually become *explicit*. Finally, human rights are eventually *institutionalised* via the law.¹³

Mahlmann argues that human rights do not merely prohibit specific actions but also secure an allocation of goods (e.g., freedom of speech) to which individuals have a right.¹⁴ He argues, further, that the principles of *justice*, *altruism*, and *respect*—and perhaps others including the *non-instrumentalisation* of individuals and principles specifying *permissible acts*—go a long way toward accounting for the moral judgments undergirding human rights.¹⁵

This brings us to moral cognition. Mahlmann seeks to ground principled moral intuitions and complex normative structures in human cognition. Moral evaluation, in *M&R*'s account, is a peculiarly *human* activity. Mahlmann writes: 'For ants, for example, the problem of self-

⁸ Ibid 49.

⁹ Ibid 79.

¹⁰ Ibid 83.

¹¹ Ibid 184.

¹² Ibid 185.

¹³ Ibid 186-188.

¹⁴ Ibid 205-206.

¹⁵ Ibid 452-453.

determination does not arise...For human beings, however, this problem does arise. Bees do not have a problem with monarchies, human beings do.’¹⁶ The moral complexity underlying human rights is a critical empirical observation about this distinctively human activity: ‘Human rights thus necessarily imply *a substantive empirical falsifiable assumption about the kind of creature human beings are.*’¹⁷

To be sure, the link between moral cognition and human rights is not just an interesting foil for Mahlman’s preceding argument. A theory of human rights ‘suffers from a potentially highly relevant blind spot if it does not concern itself with the theory of the human mind that, after all, generates this thought.’¹⁸ Mahlmann is thus concerned with the *species-specificity* of human morality. He agrees with Tomasello’s observation that, while cultural influences on moral attitudes are real, household pets exposed to the same cultural influences ‘do not thereby become moral beings...there is also absolutely no question that human children are biologically prepared for the process as well.’¹⁹

Mahlmann thus introduces the ‘mentalist’ theory of law and ethics. This approach is articulated *after* an analytical theory of morality has been constructed²⁰ because ‘no psychological or neuroscientific theory can afford any naivete about the complexity of the explanatory task at hand.’²¹ Description precedes explanation.

The mentalist theory is premised first upon the invocation of a competence-performance distinction. That is, between ‘the faculty to perform a certain cognitive task and the actual performance of this task.’²² The mentalist approach is interested in characterising *competence*, not the actual *exercise* of moral judgment in any arbitrary instance. Specifically, it seeks to determine ‘whether it is possible to identify generative principles of moral judgment specific to human moral cognition that are universal and uniform across the species—a universal moral grammar...’²³ The primary means by which this is done is through ‘considered judgments’—those which ‘are not skewed by interest, passion, errors of fact and so forth.’²⁴ The use of considered judgments reflects the understanding that humans possess a moral faculty that is in regular and complex interaction with other cognitive systems that cloud its operations.

The *acquisition* of a moral competence is a conceptually distinct matter, for which Mahlmann invokes a poverty of the stimulus argument for the moral domain. Facts about human moral competence—including the restricted range of phenomena to which moral evaluation is applied²⁵ and concepts like *ought*, *intention*, and *duty* and their associated *moral emotions*—are best

¹⁶ Ibid 273.

¹⁷ Ibid.

¹⁸ Ibid 325.

¹⁹ Michael Tomasello, *A Natural History of Human Morality* (Harvard University Press 2016) 154.

²⁰ Mahlmann, *Mind and Rights* (n 1) 402-403.

²¹ Ibid.

²² Ibid 404.

²³ Ibid.

²⁴ Ibid 403.

²⁵ Ibid 405-406.

explained in reference to their underdetermination by external stimuli; they are innate and biologically endowed.²⁶

M&R's mentalist theory of law and ethics is summarised as follows: Moral judgment is an observable phenomenon. The act of moral evaluation involves complex normative concepts that often correspond with one another (duties and claims), as they do most strikingly in the case of human rights. This form of judgment appears to be distinctively human. Thus, human beings possess a species-specific and complex capacity for moral evaluation whose use is bound up in their social and political lives. It is useful, then, to make a distinction between competence and performance, the former referring to the capacity for moral judgment and its associated concepts and the latter to the actual exercise of this capacity. This competence is targeted at a restricted range of phenomena and recruits concepts including *agency*, *ought*, *intention*, and the like. The mentalist approach seeks to characterise moral *competence*, not performance. Human rights conventions like the Universal Declaration of Human Rights (UDHR) are thus not meant to be explained by this account. Rather, the biologically endowed moral faculty 'could...turn out to be the cognitive precondition for the possibility of ultimately producing something like the Universal Declaration and the aspirations it implies.'²⁷

3. Theoretical Foundations of Universal Moral Grammar

3.1 The Central Arguments and Their Forms

So far as *M&R*'s mentalist account of moral cognition is concerned, there are two central arguments. Following Mikhail's explication,²⁸ we identify these as an *argument for a moral grammar* and an *argument from the poverty of the moral stimulus*. The first refers to the postulation that humans possess a specific kind of competence characterisable in terms of rules or principles that serve as the premises from which an unlimited number and variety of moral judgments are derived—a moral grammar. This is what Rawls described as 'a skill in judging things to be just and unjust, and in supporting these judgments by reasons...To see this it suffices to note the potentially infinite number and variety of judgments that we are prepared to make.'²⁹ The second argument refers to the notion that environmental stimuli cannot, by themselves, account for the acquisition of this moral grammar.³⁰

As Mikhail notes, these arguments respectively answer two questions: (1) What constitutes moral knowledge? and (2) How is moral knowledge acquired?³¹

²⁶ Ibid 433–436.

²⁷ Ibid 444.

²⁸ John Mikhail, *Elements of Moral Cognition: Rawls' Linguistic Analogy and the Cognitive Science of Moral and Legal Judgment* (Cambridge University Press 2011) 17.

²⁹ John Rawls, *A Theory of Justice* (Harvard University Press 1971) 46.

³⁰ Although the two arguments are conceptually distinct, the argument that individuals possess mental linguistic grammars tends to initiate the approach to UG, at which point "[t]he next question is: How did they get there?" Ray Jackendoff, *Patterns in the Mind* (Basic Books 1994) 21.

³¹ Mikhail, *Elements* (n 28) 15.

Both arguments “have the logical form of what is often called an ‘abductive’ argument...or an ‘inductive inference to the best explanation.’”³² That is, it is recognised that there ‘will be several hypotheses which might explain the evidence, so one must be able to reject all such alternative hypotheses’ before settling on a specific hypothesis.³³ This is consistent with *M&R*’s approach to a theory of moral cognition: a descriptive account of human morality, particularly within the space of human rights, is built up throughout the book. When the subject of moral cognition arises, this description is then used ‘to reconstruct the psychological mechanisms underpinning human moral judgment and investigate how they are acquired,’³⁴ while also determining, on a case-by-case basis, whether alternative approaches can account for the facts about moral psychology.

Mahlmann’s argument is that they cannot: Greene’s Dual-Process model of moral judgment fails to identify the relevance of the means-side effect distinction in the footbridge trolley problem.³⁵ It also contradicts itself by failing to account for the ‘deontological residue at the core of utilitarianism;’³⁶ Haidt’s articulation of Moral Foundations Theory fails to account for the distinction between factual, traditional morality (pre-reflective) and critically reflective ethics (i.e., considered judgments);³⁷ Haidt’s Social Intuitionist model of moral judgment, because it views moral judgments as serving the function of strategic communication, fails to identify the *idea* and *history* of human rights as bound up in reflective considerations;³⁸ and Nichols’ emphasis on statistical learning in morality acquisition cannot account for the acquisition of non-act-based moral rules (e.g., the moral relevance of certain intentions) and the concept and phenomenology of *ought*.³⁹

This (inexhaustive) list of cases reveals that Mahlmann’s form of argument is decidedly *abductive*.

First, we must flesh out exactly what kind of claim is being made in positing the existence of a universal moral grammar.

3.2 What Kind of Concept Is UMG?

It looks as though UMG is a strong claim. Is it?

On the contrary, Mendívil-Giró notes that its linguistic analog, UG, is ‘a very conservative concept.’⁴⁰ Mikhail understands UMG to be a ‘simplifying assumption, or null hypothesis, that human beings share a common moral nature.’⁴¹ Its limited nature—restrictive in how much theoretical ground UG claims for itself—is an important fact that will recur.

³² Ibid 17.

³³ Gilbert H. Harman, ‘The Inference to the Best Explanation’ (1965) 74, *The Philosophical Review* 89.

³⁴ Mahlmann, *Mind and Rights* (n 1) 402.

³⁵ Mahlmann, *Mind and Rights* (n 1) 339-340.

³⁶ Ibid 341. On Greene’s argument, see, Joshua Greene, *Moral Tribes* (Penguin Books 2013).

³⁷ Mahlmann, *Mind and Rights* (n 1) 421-422. On Haidt’s argument, see, Jonathan Haidt, *The Righteous Mind* (Vintage Books 2012).

³⁸ Haidt, *Righteous Mind* (n 37).

³⁹ Mahlmann, *Mind and Rights* (n 1) 439-443. On Nichols’ argument, see, Shaun Nichols, *Rational Rules* (Oxford University Press 2021).

⁴⁰ José-Luis Mendívil-Giró, ‘Is Universal Grammar ready for retirement? A short review of longstanding misinterpretation’ (2018) 54 *Journal of Linguistics* 860.

⁴¹ Mikhail, *Elements* (n 28) 57.

The concept of UG has its modern origins in the turn from the scientific study of language as a discipline not concerned with unobservable mental entities to studying the internal cognitive mechanisms that enable linguistic behaviour (itself reflecting the competence-performance distinction).⁴² UG, in its most basic and unchanging formulation, is ‘the theory of the requirements that human nature—human biology—imposes on the systems of linguistic knowledge which are eventually developed, that is, languages (in the sense of I-languages).’⁴³

Reiss describes UG as a *postulate*; a theoretical primitive. UG is a postulate in the sense that it allows generative linguists to link data from, say, English and Japanese by treating them each as an attained mature state of the language faculty (I-languages).⁴⁴ UG therefore refers to a real mental *object* and a *theory* about that object, thus setting up the distinction between *UG-object* (the initial state of the language faculty) and *UG-theory* (the theory of the initial state of the language faculty).⁴⁵

Similarly, then, UMG is ‘a theory of the initial state of the moral faculty...’⁴⁶ The same distinction is implied here: *UMG-object* refers to the initial state of the moral faculty whereas *UMG-theory* refers to the theory about that object.

What theoretical content, though, is included by purporting to study a cognitive *faculty* rather than, loosely speaking, a *capacity*?

The difference between *faculty* and *capacity* can be thought of as the difference between domain-*specific* cognitive systems and domain-*general* cognitive systems. Both refer to the orientation of the study of internal cognitive mechanisms but differ in whether such mechanisms are specialised for a particular form of cognition or serve general, less-defined purposes.⁴⁷ Linguistic expressions and moral judgments, on the mentalist accounts, are complex and rich phenomena—they reflect knowledge that is organised in a highly specific and intricate fashion. The cognitive systems that enable their exercise are *specialised* to these ends. The argument for a mental grammar in each domain, then, posits that individuals possess grammars consisting of rules or principles *specific to language and morality*.

Thus, UMG-theory is the theory of the *initial state* of a *domain-specific cognitive system* for moral judgment characterisable in terms of *rules and principles* whose contours are imposed by *human biology* (UMG-object).

How, though, do *rules* and *principles* enter into UMG-theory?

The invocation of rules and principles is part of a *descriptively* adequate account of the sense of justice. Indeed, that individuals exhibit complex moral judgments that appear to rely on ‘structure-

⁴² Ibid 18-19.

⁴³ Mendiñil-Giró, *Universal Grammar* (n 40) 861.

⁴⁴ See generally, Charles Reiss, ‘Research Methods in Armchair Linguistics’ in Gabe Dupre, Ryan Nefdt and Kate H. Stanton (eds.), *Oxford Handbook of Philosophy of Linguistics* (Oxford University Press, In Press).

⁴⁵ Ibid 6-9.

⁴⁶ Mikhail, *Elements* (n 28) 15.

⁴⁷ Steven Pinker, *The Language Instinct* (HarperCollins 2007) 4-5.

dependent rules’ in part comprise the argument that they possess mental grammars, but this is ‘logically prior’ to an explanatory framework that allows researchers ‘to ask *whether* they are innate.’⁴⁸ This invocation effectively says: acquisition notwithstanding, individuals intuitively *know* certain moral rules or principles.

The structure of *M&R* reflects this, and harkens back to the abductive nature of the central arguments: For example, Mahlmann takes issue with Nichols’ statistical learning account by noting that ‘the acquirendum seems to be of a different and much more complex nature than is allowed for in the doing/allowing distinction,’⁴⁹ but only *after* constructing an analytical theory of morality through the lens of human rights, where the complexity of this capacity for moral evaluation becomes apparent.

Furthermore, the word ‘grammar’ in ‘Universal Grammar’ can be confusing in the linguistic analogy. As Dékány explains, “grammar” in this sense does not refer to phoneme, morpheme, and word order—it is not the ordering of the words on this page. Rather, it ‘encodes the general organisational principles of human language and makes no reference to surface order.’⁵⁰ Generative grammar is aimed at uncovering invariant commonalities in human languages.⁵¹

Remember, UG takes its object of study to be *internal* to the mind, meaning the ordering of the words on this page will consist only of clues that help researchers uncover the rules or principles responsible for their organisation across all languages. Thus, given that UG is aimed at studying the *initial state* of the language faculty, the principles used to explain it are pursued in this *cognitive* context—not within the context of sociocultural constructions like English or Japanese (E-languages) but within the context of an internal cognitive system that determines the shape of all natural languages (I-language). The same holds for UMG and the separation of internal, I-morality and external, E-moralities (cultural systems of morality).

To be clear, the claim that an aspect of the mind is internal and domain-specific is *not* tantamount to claiming that this capacity is constituted by a mental *grammar*. The individual’s ability to construct, re-construct, and understand words and sentences in novel combinations they have never previously encountered without an apparent limit on this ability, and to provide a *structural analysis* of these components, comprise the relevant phenomena for which a mental grammar accounts.⁵² Even attributing to someone the restrictive maximum capacity to construct twenty-word sentences, that individual could produce, in principle, at least 10^{20} sentences, or a one followed by twenty zeros, as Pinker illustrates.⁵³ (The preceding sentence alone is 32 words.)

A similar case holds for morality, wherein intuitive moral judgments are produced in response to situations that bear little resemblance in that ‘[e]ach is a potentially brand new combination of

⁴⁸ Mikhail, *Elements* (n 28) 82.

⁴⁹ Mahlmann, *Mind and Rights* (n 1) 440.

⁵⁰ Éva Dékány, ‘Foundations of generative linguistics’ (2019) 66 *Acta Linguistica Academia* 318.

⁵¹ Ibid. As for the refutability of UG, the cognitive turn is crucial: to refute UG is not to show that human languages exhibit diversity in surface word order properties (properties of E-languages; typological universals), but instead to show that no *organisational* principles can be shown to be present in all human languages. Ibid 321-322.

⁵² Mikhail, *Elements* (n 28) 43-46.

⁵³ Pinker, *Language Instinct* (n 47) 85-86.

agents, acts, events, and circumstances, occurring for the first time in the history of the universe...'⁵⁴ A mental grammar in the domain of morality, then, is a distinct phenomenon.

A brief historical note: Humboldt's phrase popularised by Chomsky – 'infinite use of finite means' – only became available to linguists in the early-to-mid twentieth century as a phenomenon amenable to scientific study because of the establishment of *recursive function theory* (now called *computability theory*), thereby postulating a computational system capable of such infinite generativity from finite means (UG-object). The system is tasked with the unbounded generation of structured expressions.⁵⁵ UG is the theory of *this* object. UMG, at the most basic level, holds similar characteristics.

3.3 Internalism and Diversity

The foregoing remarks are critical for UMG and human rights. Because this claim is internally oriented and is designed to explain observable behaviour, the *theory* of the moral *must be consistent with observed behaviour*. Whatever *diversity* in E-moralities exists is, and must be, a manifestation of what UMG-object permits the human being to morally perceive.⁵⁶

The conceptual significance of UMG's relationship with human rights conventions like the UDHR cannot be properly grasped unless the I-morality/E-morality distinction is articulated clearly and moral diversity is not understood to be in tension with an innate moral grammar.⁵⁷ Mahlmann's emphasis on *reflective* morality and the clarification of ethical concepts in moral problems echoes this, giving 'reason to believe that under the surface of apparently insurmountable moral disagreement, there may be a deep, shared structure of common moral principles.'⁵⁸ My interpretation of *M&R*, to borrow my own phrasing, is that 'universal human rights norms are *social constructs* which are not inevitable, but preferentially encapsulate the internal logic of individuals' moral nature.'⁵⁹ Not inevitable yet structured in advance of use.

3.4 Domain-General to Domain-Specific

Nevertheless, disagreements about UG frequently turn on the domain-specific and domain-general distinction.

For example, Chomsky will frequently summarise the broad contours of UG by contrasting his granddaughter with a rock or a rabbit, noting how only one—his granddaughter—appropriately selects the correct linguistic data from her environment and reflexively acquires a language during development.⁶⁰ Although all three can be exposed to the same data, only one identifies relevant data and acquires a language. The implication is that language must have a biological basis in humans—it is species-specific and, at least implicitly in these remarks, domain-specific.

⁵⁴ Mikhail, *Elements* (n 28) 48.

⁵⁵ David J. Lobina, 'What linguists are talking about when talking about...' (2014) 45 *Language Sciences* 2.

⁵⁶ Susan Dwyer, 'Moral Dumbfounding and the Linguistic Analogy: Methodological Implications for the Study of Moral Judgment' (2009) 24 *Mind & Language* 283.

⁵⁷ Carchidi, *Nature of morals* (n 2) 73-75.

⁵⁸ Mahlmann, *Mind and Rights* (n 1) 429.

⁵⁹ Carchidi, *Nature of morals* (n 2) 74.

⁶⁰ See, e.g., Noam Chomsky, *The Architecture of Language* (Oxford University Press 2000) 50.

Dąbrowska argues that ‘*nobody* disputes this...The crucial question is whether the relevant knowledge or abilities are language-specific or whether they can be attributed to more general cognitive processes—and this is far from clear.’⁶¹ In that vein, Tomasello argues that the *usage-based* approach evades the problem of connecting a language in the child’s environment to Universal Grammar because the latter, he argues, does not exist. Instead, ‘more regular and rule-based constructions of a language’ are acquired through learning in the same fundamental fashion that underpins the ‘learning of all complex cognitive activities...they then construct abstract categories and schemas out of the concrete things they have learned.’⁶² Thus, language construction by children is carried out through ‘general cognitive processes falling under two broad categories: (1) intention-reading...and (2) pattern-finding...’⁶³

The usage-based approach does not deny the role of internal cognitive mechanisms nor even the biological basis of language. What it denies is their specificity to language.

Both Chomsky and Tomasello would agree that human beings have a ‘biological preparedness’⁶⁴ for the acquisition of species-specific capacities. What, then, is their actual disagreement? And why does it seem that UG-theory simply *assumes* the existence of UG-object?

Mendívil-Giró attributes this seemingly impassable debate to a dispute over the internal nature of language as a cognitive system (I-language) and the diversity of languages as they manifest in human affairs (E-languages): the ‘idea that there is a biological bias for language seems impossible to be distinguished from the idea that this biological bias is specifically linguistic. Hence, the debate is in itself sterile.’⁶⁵ The locus of the disagreement is divergent conceptions of language in scientific inquiry, with critics of UG interested in language as ‘basically an object external to the mind.’⁶⁶

Mendívil-Giró further explains: the goal of UG-theory has always been unification with the natural sciences, which in turn requires a ‘reduction of what is specifically linguistic to elementary entities which are not, by definition, specifically linguistic.’⁶⁷ Any division made between an ‘emergent’ linguistic mental grammar and an ‘innate’ linguistic mental grammar therefore misses the point, as the dichotomy is false: the only way for language to arise is through the arrangement of non-linguistic elements into linguistic entities;⁶⁸ the ‘specifically linguistic’ component of human biology that UG seeks to explain is the ‘initial formulation of the properties of such entities.’⁶⁹ This is the leap from non-linguistic to linguistic; general to specific.

⁶¹ Ewa Dąbrowska, ‘What exactly is Universal Grammar, and has anyone seen it?’ (2015) 6 *Frontiers in Psychology* 3.

⁶² Michael Tomasello, ‘Acquiring Linguistic Constructions’ in Deanna Kuhn and Robert Siegler (eds.), *Handbook of Child Psychology*, vol 2, (6th edn, John Wiley & Sons 2006) 258.

⁶³ Ibid.

⁶⁴ Michael Tomasello, *Becoming Human* (Harvard University Press 2019) 301.

⁶⁵ Mendívil-Giró, *Universal Grammar* (n 40) 862-863.

⁶⁶ Ibid 863.

⁶⁷ Ibid 864.

⁶⁸ Ibid 864-865.

⁶⁹ Ibid 866.

Chomsky's response to the dispute follows this reasoning. In response to Enfield's remark that 'language is entirely grounded in a constellation of cognitive capacities that each—taken separately—has other functions as well,'⁷⁰ Chomsky says:

'Note that if something like this turned out to be true, it would have no direct bearing on the richness of [Universal Grammar], which would then have to account for the species-specific ways in which these capacities form a "constellation" in humans, dissociated from their other functions.'⁷¹

General cognitive capacities cannot make the leap from non-linguistic to linguistic if there is no specific component of human biology that allows for the linguistic orientation of this 'constellation' of capacities.

Mahlmann employs this reasoning,⁷² particularly in his response to Tomasello that new forms of human cooperation *presuppose* the relevant cognitive capacities. To the point, sentiments like sympathy and fairness do not account for the fact that moral judgments can have a *deontic* status and be assigned to the *intentions* of individuals.⁷³ This evolutionary account, Mahlmann continues, neglects the taken-for-granted notion of *legitimacy* in human social organisation and the concept of a *moral ought* that undergirds self-regulation.⁷⁴ Each of these points relates to the domain-specificity of moral judgment—its specialised, not general, orientation.

This reasoning matters a great deal for any proposed relationship between statistical learning (machine or otherwise) and belief in human rights attitudes. An approach to moral psychology that does not take the 'intricate'⁷⁵ world of normative reasoning seriously risks breaking up the capacity for moral evaluation into a network of domain-general cognitive operations that independently hold no moral quality. Analogising from Chomsky in the linguistic context,⁷⁶ there is a sense in which this approach simply denies that morality, in the cognitive domain, *exists*.

Yet, the history and idea of human rights do not make scientific sense unless the moral domain that undergirds their sincere promotion is recognised to *exist*. I interpret Mahlmann's response to Nichols' statistical learning argument to be an affirmation of morality's existence, in this sense, with emphasis on its internal and domain-specific nature. Mahlmann's argument concerning the leap from non-moral concepts to moral concepts—*no matter the quantity of examples of the latter*—is of particular importance:

'[C]ognitive access to basic elements of the moral world is the precondition for understanding what moral judgment and explicit norms are about in the first place. You need to know what *ought* means, for instance, before you can grasp what you ought to do, as Hume already argued. Statistical learning is unable to bridge this gap. No quantity of

⁷⁰ Nick J. Enfield, 'Without social context?' (2010) 329 *Science* 1600.

⁷¹ Noam Chomsky, 'Problems of projection' (2013) 130 *Lingua* 34.

⁷² Mahlmann, *Mind and Rights* (n 1) 371-373.

⁷³ *Ibid* 378.

⁷⁴ *Ibid* 380-381.

⁷⁵ *Ibid* 324.

⁷⁶ See, Chomsky, *Problems of projection* (n 71) 34.

references to ought by others will tell you what ought means from a first-person perspective...’⁷⁷

As I understand this argument, it relates directly to the cognitive *existence* of morality: to know that one *ought* to do something assumes that one can access an ‘identifiable cognitive domain.’⁷⁸ Mahlmann’s position here is in effect barring the leap from non-moral to moral entities without pre-existing, internal resources of a domain-specific nature.⁷⁹

Mahlmann’s argument about the leap from non-moral to moral is explicitly grounded in poverty of the stimulus considerations—individuals *know* or *understand* what it means that they *ought* to do something, and *how* they know this relies on pre-specified biological resources. This echoes a broader point about the complexity of moral judgment’s normative structure and the principles that individuals appear to acquire, particularly in *M&R*’s discussion of the relationship between *intentionality* and *duty* in the context of the doing/allowing distinction⁸⁰—a point made, again, *after* an analytical theory of morality is constructed.

With this in mind, we turn to statistical machine learning and its challenges to nativism.

4. The Large Language Model Challenges

Rather than the ‘humble and familiar forms of inference’⁸¹ often discussed in statistical learning, we are principally concerned with statistical learning as embodied in machine learning models. There is overlap, to be sure, as shown above and below.

To be clear, LLMs specifically, much as they have sparked this intersection of interest, are not *direct* challenges to UMG, lacking equivalencies for concepts like *intentions* and exhibiting shifts in “personality” incomparable to that of humans when the latter morally evaluate.⁸²

Still, it is *how* LLMs are being leveraged in linguistics that *is* directly relevant to how the mentalist theory could be challenged via computational modelling.

4.1 LLMs As Theories and LLMs as Proxies

The LLM-driven challenge to UG has different variants, stronger and weaker in their claims. Ziv et al.⁸³ categorise the claims as either *LLM Theory*—in which the LLM is a theory *in its own*

⁷⁷ Mahlmann, *Mind and Rights* (n 1) 439.

⁷⁸ Mikhail, *Elements* (n 28) 95.

⁷⁹ Contrasts exist. For an argument that morality should instead be seen as domain-general problem-solving, see Mark Johnson, ‘There is no moral faculty’ (2012) 25 *Philosophical Psychology* 426.

⁸⁰ Mahlmann, *Mind and Rights* (n 1) 440.

⁸¹ Nichols, *Rational Rules* (n 39) 15.

⁸² Madeline G. Reincke et al., ‘The puzzle of evaluating moral cognition in artificial agents’ (2023) 47 *Cognitive Science* 1-7.

⁸³ Imry Ziv, Nur Lan, Emmanuel Chemla and Roni Katzir, ‘Large language models as proxies for theories of human linguistic cognition’ [2025] arXiv 1-3.

right—and the *Proxy View*—in which LLMs are proxies for theories explicitly developed elsewhere.⁸⁴ Most proponents appear to fall into the latter category.

Piantadosi’s critique of Chomsky’s approach to language hinges on the idea that LLMs, by virtue of their ability to ‘develop representations of key structures and dependencies’ therefore ‘should be treated as bona fide linguistic theories.’⁸⁵ LLMs are theories *in themselves* that refute the generative approach to language. Ambridge and Blything take a similarly forward posture, arguing that LLMs are ‘better theories than any other that have been proposed.’⁸⁶ These are LLM Theory positions.

Contreras Kallens et al. argue that LLMs vindicate the promise of earlier, shallower connectionist models by demonstrating that the ability to generate grammatical language—understanding and other characteristics like intelligence notwithstanding—*can* be attained via statistical learning at scale.⁸⁷ They write: LLMs ‘can be viewed as a usage-based answer to the [poverty of the stimulus] argument—at least when it comes to the production of grammatical language.’⁸⁸ This is the Proxy View.

Others are enthusiastic, but more amenable to compromise. Futrell and Mahowald argue that theoretical linguistics and LLMs are *mutually beneficial* and should aid one another’s developments.⁸⁹ Griffiths et al. similarly argue that neural networks generally (though, the angle toward transformers is clear) exhibit abilities to combine ideas, produce novelty, and learn with facility that may offer important, if limited, insight into human cognition.⁹⁰ These are more moderate instances of the Proxy View.

The common theme is: LLMs produce novel grammatical expressions without innate linguistic constraints, instead achieving their performance by training via next-token (next-word) prediction on text-data. As a result, LLMs are either a *theory* or an *existence proof* that human language can be acquired primarily through data. Therefore, if one is weighing competing hypotheses for human language, an inference to the *best* explanation is not that which postulates an innate linguistic grammar but that which attributes *something closer* to the operations undertaken by a domain-general, non-modular system like a transformer.

Like the presentation of UMG in *M&R*, LLMs are evaluated via *inference to the best explanation*; abduction. In that vein, I provide some less appreciated disanalogies between LLMs

⁸⁴ In moral psychology, a distinction close to this exists, namely Nichols’ distinction between a “how possible” account (how moral learning *might* proceed) and a “how actual” account (how moral learning *actually* proceeds), Nichols, *Rational Rules* (n 39) 153.

⁸⁵ Steven T. Piantadosi, ‘Modern language models refute Chomsky’s approach to language’ in Edward Gibson and Moshe Poliak (eds.), *From Fieldwork to Linguistic Theory* (Berlin Language Science Press 2024) 360.

⁸⁶ Ben Ambridge and Liam Blything, ‘Large language models are better than theoretical linguists at theoretical linguistics’ (2024) 50 *Theoretical Linguistics* 34.

⁸⁷ Pablo Contreras Kallens, Ross D. Kristensen-McLachlan and Morten H. Christiansen, ‘Large language models demonstrate the potential of statistical learning in language’ (2023) 47 *Cognitive Science* 1-6.

⁸⁸ *Ibid* 3.

⁸⁹ Richard Futrell and Kyle Mahowald, ‘How linguistics learned to stop worrying and love the language models’ [2025] *Behavioural and Brain Sciences* 1-98.

⁹⁰ Thomas L. Griffiths et al., ‘Whither symbols in the era of advanced neural networks?’ [2025] arXiv.

and humans below.

4.2 Disanalogies Between LLMs and Humans

First, LLMs process text through ‘tokens,’ which are words or parts of words. Each token has a numerical value (e.g., ‘dog’ may be two tokens, each represented by a number). During pre-training, it predicts the next token of a partially masked sequence of tokens. The model learns the statistical correlations between these tokens during pre-training based on a training dataset comprised of tokenised text-data generated by humans.⁹¹ The model therefore does not ‘read’ or ‘process’ natural language text as *words*, a basic linguistic unit for humans.⁹²

This point is more important than it may appear. Consider the following:

The list of behaviors of which knowledge of language purportedly consists has to rely on notions like ‘utterance’ and ‘word.’ But what is a word? What is an utterance? These notions are already quite abstract. Even more abstract is the notion ‘sentence’. Chomsky has been and continues to be criticized for positing such abstract notions as transformations and structures, but the big leap is what everyone takes for granted. It's widely assumed that the big step is going from sentence to transformation, but this in fact isn't a significant leap. The big step is going from ‘noise’ to ‘word.’⁹³

If is the ‘big step’ is moving from ‘noise’ to ‘word’, then we may conceive of LLMs as effectively circumventing it through tokenisation. Lasnik et al.’s remark can be interpreted as assigning importance to the leap from domain-general to domain-specific; non-linguistic to linguistic.

Here, LLMs do not confront the same challenge as a child. LLMs do not pre-train on text-data *per se*. Rather, they pre-train on tokenised text-data. Thus, for an LLM, there is no ‘noise’ in its environment from which it must select relevant linguistic data. The data on which it trains is the only data in the model’s ‘environment’. The linguistic data is effectively circumvented through pre-training on tokenised text-based utterances.⁹⁴

Second, though the data LLMs are trained on is unstructured (which *is* narrowly analogous to human learning), many of these data are rich linguistic texts of ample diversity (think: academic textbooks, novels, essays, and so forth). It is as if we were to suggest that human children read *A Tale of Two Cities* before learning to speak. Thus, the LLM’s learning is *sharply disanalogous* to the human child’s learning *sequence*, ‘reading’ text before acquiring an ability to speak.

⁹¹ When interacting with an LLM-based chatbot, the tokens outputted are decoded into natural language characters for the human’s interpretability.

⁹² Stela Manova, ‘ChatGPT and linguistic theory, with a focus on morphology’, in José-Luis Mendivil-Giró (ed.), *Artificial Knowledge of Language* (Vernon Press, In Press).

⁹³ Howard Lasnik, Marcela Depiate and Arthur Stepanov, *Syntactic Structures Revisited* (The MIT Press 2000) 3.

⁹⁴ Relatedly, see, Veno Volenec and Charles Reiss, ‘Adopting large language models as a theory of language does refute Chomsky (but not like you think)’ (2025) 22 SKASE Journal of Theoretical Linguistics 5-6.

Third, LLMs do not confront the same challenge as a child; they *do* have a learning objective, but the *means* by which this objective is pre-scaffolded by human researchers ensures that the heavy cognitive lift of moving from ‘noise’ to ‘language’ is circumvented.

LLMs are built via statistical learning algorithms. As Yang noted amid a burgeoning debate about statistical learning in human language acquisition, any (effective) learning algorithm must appropriately represent the relevant learning data. Thus, the claim that infants statistically track information ‘must presuppose that children know *what kind* of statistical information to keep track of.’⁹⁵ Yang’s reasoning turns on the range of *possible* correlations of linguistic data an infant could track, indicating that this range is *constrained* from the get-go by an innate, domain-specific endowment (i.e., UG-object). The constraints ensure that the child’s learning operates over *specific aspects* of the input.⁹⁶

LLMs partly circumvent this challenge. Specifically, the human-given training objective *constrains* the LLM to a particular set of statistical relationships. This is not to suggest that the human knows in advance which correlations will be made and how these will manifest in the model’s eventual outputs in production. It merely implies that humans provide the training objective (and the data) which constrains the model’s learning trajectory. The same cannot be said of human children where a language will be acquired whether explicit instruction to follow a particular learning trajectory is provided or not.⁹⁷

An interlocutor may argue that human engineers *substitute* the role of genetics in computational modelling, instantiating a learning objective in LLMs prior to training that is ‘innate’ but domain-general. In the archetypal case of next-token prediction in LLMs, although it is provided from the ‘outside’, this fact does not detract from its domain-generality (agnostic to the training dataset, with no language-specific principles constraining it). One must be permitted to ‘set in motion’ a model’s learning to test specific learning objectives!

This point appears to rebut Yang’s argument that domain-specific constraints are necessary. LLMs, the interlocutor argues, acquire language merely through exposure to linguistic data. Thus, an *existence proof* of domain-general language acquisition is provided by LLMs.

However, compelling as this point is, it depends on the LLM’s learning being sufficiently analogous to that of a human child. There is reason to doubt this. Among them, the use of statistical learning via next-token prediction is constrained *by* human engineers by virtue of training on a *closed-set of language-specific data over which this learning objective operates*. The model’s ability to learn a human structure depends on the training dataset having an ‘answer’ to the question of which token comes next in a sequence; this answer is provided by the training dataset, *restricted* to products generated by humans already in possession of a linguistic competence, unstructured in nature but *curated on the model’s behalf*.⁹⁸

⁹⁵ Charles D. Yang, ‘Universal Grammar, statistics or both?’ (2004) 8 Trends in Cognitive Sciences 452.

⁹⁶ Ibid 455.

⁹⁷ Moreover, humans universally settle on certain linguistic categories, including ordered phonological processes, morphemes, and universal syllable structure, Volenec and Reiss (n 94) 7-8.

⁹⁸ One related way of illustrating the disanalogies between human and LLM learning is to explicate the *epistemic path* taken by LLMs to produce *judgments* (or answers) and how it relates to human learning. See, Walter

LLMs therefore partly circumvent the challenge highlighted by Yang in that statistical learning is used only *after* the model's hypothesis space has been *somewhat restricted* by human engineers through a language-specific training dataset. Statistical learning does not bridge the gap between 'noise' and 'language', so much as the stage has been set for it operate on. Note that this training regime is fully consistent with Mahlmann's own claim about statistical learning: '[C]ognitive access to basic elements of the moral world is the precondition for understanding what moral judgment and explicit norms are about in the first place.'⁹⁹

The object of study in generative linguistics is not strings of tokens. Rather, it is the system of unbounded generativity of form/meaning pairs; language is *meaning with sound*.¹⁰⁰ This computational system must have its own constraints. Any use of computational models as theories or proxies must therefore evidence comparable constraints.

The transformer may be *differently constrained* than human minds, given both the kinds and quantities of correlations tracked through next-token prediction on a massive dataset during pre-training. This was Chomsky's point in claiming that LLMs could learn *impossible* languages with equal facility as *possible* languages.¹⁰¹

The claim of equal facility was made prematurely by Chomsky. Nonetheless, results have trended in this direction. Kallini et al.¹⁰² initially found that GPT-2 models trained on *possible* (attested) languages learned more efficiently, indicating they are constrained similarly to human acquisition.¹⁰³ However, Ziv et al. extend Kallini et al.'s test of impossible (unattested) languages by training a transformer-based LLM on the latter's original perturbed versions of languages and added an *additional perturbation* across English, Italian, Russian, and Hebrew. Several of the unattested languages were *easier* for the transformer to learn using the same metric,¹⁰⁴ indicating divergent constraints.

Indeed, the differently constrained nature of transformers is also evidenced by distinctive failure modes in LLMs. David Lobina argues that *adversarial attacks*—perturbations to the model's input data with the aim of interrupting its operations—that are effective against LLMs, but not against humans, is revelatory: the model is susceptible to such attacks *because* it does not regulate form/meaning pairs, instead modelling text-data.¹⁰⁵ Inference to the best explanation requires that

Quattrocioni, Valerio Capraro and Matjaž Perc, 'Epistemological fault lines between human and artificial intelligence' [2025] arXiv.

⁹⁹ Mahlmann, *Mind and Rights* (n 1) 439.

¹⁰⁰ Elliot Murphy, 'Beyond prediction: comments on the format of natural intelligence' [2025] Cognitive Neuroscience 2.

¹⁰¹ Noam Chomsky, Ian Roberts and Jeffrey Watumull, 'The False Promise of ChatGPT' *The New York Times* (8 March 2023) <www.nytimes.com/2023/03/08/opinion/noam-chomsky-chatgpt-ai.html> accessed 6 November 2025.

¹⁰² Julie Kallini, Isabel Papadimitriou, Richard Futrell, Kyle Mahowald and Christopher Potts, 'Mission: Impossible languages' [2024] arXiv.

¹⁰³ That impossible languages were learned by transformers *at all*, albeit less efficiently, is a (comparatively less extreme) divergence in its own right.

¹⁰⁴ Ziv et al. (n 83) 15-20.

¹⁰⁵ David J. Lobina, 'Language models as function approximators of text data' in José-Luis Mendiñil-Giró (ed), *Artificial Knowledge of Language* (Vernon Press, In Press).

proponents of LLMs as models of human language formulate a *better* explanation for LLMs' continuing susceptibility to such attacks than the claim that they instead model text-data.

Another way to conceive of the target of UG-theory is therefore a strong generative capacity: linguistic knowledge that enables 'a grammar to generate *structural descriptions* for a language. A structural description is an analysis of a sentence in terms of phonology, morphology and syntax.'¹⁰⁶

M&R travels within this tradition. Humans do not merely produce statements about the rightness or wrongness of actions; they produce moral *judgments* that reflect a rich *knowledge* of morality. They can engage in 'an analysis of structural elements of the action (agency, patients of action, intentions and their kind and object, relations of equality, etc.).'¹⁰⁷ A moral evaluation takes components such as intentions or actions and imposes on the arrangements of facts in the given case principles that then give a 'deontic status' to the relevant component.¹⁰⁸ Indeed, what Mahlmann calls 'reflective' morality is an encapsulation of *strong generative capacity* in the moral domain— '[o]nly critical reflection on morality, based on considered judgments, will tell us what human morality ultimately is about...'¹⁰⁹

5. The Small-Data Challenge

Our remarks thus far target LLMs pre-trained on massive datasets. The goal for some, however, is to construct a computational model that can acquire a language with 'developmentally plausible amounts of data,'¹¹⁰ thereby evading the charge of inhuman data exposure. A recent experiment using a small machine learning model claims to make a step in this direction, with its authors suggesting it holds serious implications for nativist theories of language acquisition.

Vong et al. observe how a long-standing problem in philosophy and cognitive science is how children who learn new words are confronted with a 'potentially infinite set of candidate meanings.'¹¹¹ A simple utterance—e.g., 'Do you want the teddy bear?'—poses to the child the problem of linking the words 'teddy bear' with an object in their environment. How the child manages to quickly learn to associate words with objects correctly has given rise to debates over whether human children require domain-*specific* and innate constraints or can achieve their word-referent mappings with domain-*general* mechanisms.

Vong et al. thus ask:

If a model could perceive the world through a child's eyes and ears, would it need strong inductive biases or additional cognitive capacities to get word learning underway? Or

¹⁰⁶ John Collins, Chomsky: *A guide for the perplexed* (Continuum International Publishing Group 2008) 50.

¹⁰⁷ Mahlmann, *Mind and Rights* (n 1) 346.

¹⁰⁸ Ibid 416.

¹⁰⁹ Ibid 421.

¹¹⁰ Piantadosi, *Language Models* (n 85) 367.

¹¹¹ Wai Keen Vong, Wentao Wang, A. Emin Orhan and Brenden M. Lake, 'Grounded language acquisition through the eyes and ears of a single child' (2024) 383 *Science* 504.

would a simpler account driven by associative learning, in conjunction with feature-based representation learning, suffice?¹¹²

The researchers test this in a striking way: by attaching a lightweight camera to the head of a child, beginning at 6 months old and ending at 25 months old (1.5 years overall), totalling over 600,000 video frames and 37,500 text transcriptions of utterances (the camera captured roughly 1% of the child's waking hours). They brought in a multimodal machine learning model—called Child's View for Contrastive Learning (CVCL)—which is designed to implement one form of associative learning 'that is cross-situational, tracking the co-occurrences between words and possible visual referents to determine their mappings.'¹¹³ The model leverages both representation learning¹¹⁴ and associative learning and its objective is to coordinate a vision encoder and a language encoder. It is trained via self-supervision.¹¹⁵

After training, CVCL was given two tasks, each of which involved matching a word with an image. The first task had the model select from images in the original dataset and the second task gave the model the option of picking from images *not* within the original dataset and were thus novel to CVCL. The model scored 61.6% accuracy on the first task (the original dataset) and 34.7% accuracy on the second (the novel dataset)—both are above the level of chance, which is 25%.¹¹⁶ While the model's performance drops when confronted with novel images, the researchers maintain that this success, though limited, augurs future techniques that may well demonstrate higher performance.

The core significance of these results, they argue, lies in challenging the perceived difficulty of early language acquisition by nativist theories, instead highlighting how 'learnability with minimal ingredients shows how representation learning and associative, cross-situational mechanisms are sufficient to acquire word-referent mappings from one child's first-person experiences.'¹¹⁷ CVCL is thus used as a strong *proxy* for existing theories.

5.1 Does Small-Data Machine Learning Threaten Nativism?

CVCL is interesting, though the experiment is deficient when put to the end of arguing against nativism.

First, the data on which CVCL is trained was collected *by* the child and *for* the model. The model is piggybacking off the data collection conducted by the child who navigates her environment *autonomously*.

¹¹² Ibid.

¹¹³ Ibid.

¹¹⁴ Most basically, representation learning in machine learning is the process by which a model learns representations that allow it to identify specific features in datasets.

¹¹⁵ Vong et al. (n 111) 504.

¹¹⁶ Ibid 506-508.

¹¹⁷ Ibid 509.

As Yiu et al. argue, among the core differences between children and models is that ‘children are curious, active, self-supervised, and intrinsically motivated.’¹¹⁸ This disanalogy applies as well to CVCL, where the model does not autonomously seek out or find relevant data in the environment, but instead takes the *child’s self-directed behaviour*—who is already in possession of the cognitive capacities Vong et al. are interested in studying—as its source of training data.

To be sure, neural networks like CVCL can be said to discriminate among their training data, assigning more importance to some data points than others, perhaps indicating that the algorithm is operating like the child.¹¹⁹

There are two problems with this. One is that the child *seeks out* the relevant data in their environment *without* an explicit training objective; a parent may direct a child in some contexts, but the child does not require explicit instruction to seek out—give due weight to—the data relevant for word-referent mappings. CVCL trains only because it is directed; there is no autonomy, in this sense.

The other problem is that, while one may plausibly argue that the updating of weights in a neural network during training is a narrowly autonomous activity,¹²⁰ this is quite a different kind of activity than what the human child performs. The child does not require that it be closed off from all other factors; there is simply environmental exposure from which the child acquires a language. The neural network would not update its weights were it not provided with a training objective; it functions, but it does not act.¹²¹

Experiments like CVCL are intuitively appealing because they appear to resolve the problem captured by James’ description of a human baby’s world as a ‘blooming, buzzing confusion’.¹²² Yet, the ‘special discriminative act’¹²³ James describes in the same passage is usefully conceived as the child’s ability, when confronted by a world of mere data, to move from ‘noise’ to ‘language’. There is thus a relevant distinction to be made between training a model on *unstructured data* and *collecting data in an unstructured environment* such that an ability to associate objects and words and map forms with meanings over an unbounded range is acquired.

Re-consider, then, Vong et al.’s question: ‘If a model could perceive the world through a child’s eyes and ears, would it need strong inductive biases or additional cognitive capacities to get word learning underway?’¹²⁴ Yet, the model does have at least one strong inductive bias: *the child’s behaviour*.

Second, CVCL is trained and tested on video *frames*. This is unlike a human child who navigates autonomously in a *fluid* and *dynamic* environment without the benefit of static and curated data.

¹¹⁸ Eunice Yiu, Eliza Kosoy and Alison Gopnik, ‘Transmission versus truth, imitation versus innovation’ (2024) 19 *Perspectives on Psychological Science* 880.

¹¹⁹ I am grateful to Tom McCoy for raising this objection in the context of the co-authored paper referenced above (Griffiths et al. (n 90)).

¹²⁰ As Gubelmann does in the case of LLMs. Reto Gubelmann, ‘Large language models, agency, and why speech acts are beyond them (for now) – A Kantian-Cum-Pragmatist case’ (2024) 37 *Philosophy & Technology* 3-9.

¹²¹ *Ibid* 1-24.

¹²² William James, *The Principles of Psychology, Vol. 1* (Henry Holt and Company 1918) 488.

¹²³ *Ibid*.

¹²⁴ Vong et al. (n 111) 504.

In contrast, CVCL deals with linguistic data that have been isolated and frozen, somewhat akin in this regard to the use of tokenisation in transformers.

Third, the researchers use their experiment's results to question theories of linguistic nativism, yet such an experiment merely tests for *associations* of words and meanings. This is a much different matter than testing for a *mental grammar* (an I-language).

Human language contains 'an inventory of elements like words that can be combined and recombined in limitless new ways to express new messages,'¹²⁵ expressions that are immediately understood by others who have 'never heard or spoken' them in their personal history.¹²⁶ Humans do not merely associate words with objects,¹²⁷ but rather construct new sentences that have principled structures and which can be used to express new *thoughts* and *ideas* over an unbounded range.

The reasoning here resembles a criticism that Mikhail raised about Nichols' *Rational Rules*: in the case of moral grammar, there is a gap between the *elements* of moral cognition (e.g., moral concepts and distinctions like the means/side effect distinction) and the recruitment of *deontic rules* that use these elements to construct intuitive and novel moral judgments over an unbounded range.¹²⁸ In Mikhail's account, moral rules are the premises from which moral judgments are derived—the *argument for a moral grammar* seeks to pursue an explanation through the confines of a mental grammar. Mahlmann echoes this in *M&R* by noting that the mentalist approach not only identifies that certain facts trigger the use of principles that determine moral evaluation, but also seeks to 'determine *how* moral judgments supervene over facts (not just that they do) and, second, which facts are morally relevant in the first place...' ¹²⁹ Principles as part of a moral grammar 'specify how moral judgments supervene over facts...' ¹³⁰

Although Vong et al. are concerned with language acquisition, CVCL only provides a very limited test case of how words and objects might relate; a proxy. Nor does CVCL escape some of the fundamental problems faced by LLMs. Although it differs in that it is not trained on massive quantities of data, the data it *is* trained on is autonomously sought out by the *child*, being frozen and static in the form of video frames. In any event, CVCL only acquires above-chance word-referent mappings—a grammar is a different matter.

6. The Voluntary Action Problem

There is another problem: voluntary human language use.

Human beings are not input-output mechanisms. They *use* their language in ways that cannot meaningfully be said to be *caused* by situations nor internal states, often expressing novel

¹²⁵ Jackendoff (n 30) 10.

¹²⁶ Ibid 11.

¹²⁷ Nichols explains how the poverty of the stimulus argument is framed in terms of acquiring a sophisticated state of grammatical knowledge, *assuming* the child possesses some knowledge of words and objects, Nichols, *Rational Rules* (n 39) 142-143.

¹²⁸ See generally, John Mikhail, 'Book review' (2022) 131 *The Philosophical Review* 399-403.

¹²⁹ Mahlmann, *Mind and Rights* (n 1) 411.

¹³⁰ Ibid 457.

utterances, yet these utterances are nonetheless *appropriate* to the situation in that they align with the mental states of others. Descartes and later Cartesians recognised this as the problem of being able ‘to respond appropriately to the input while still being ‘detached’ from that input (‘stimulus-free’) [*sic*].’¹³¹

Ordinary language use is *stimulus-free, unbounded, yet appropriate*. While Chomsky effectively argued against Skinner’s notion of *external* stimulus-control,¹³² the later and more important point is often missed: there is no meaningful sense in which causality can account for language use even from the perspective of *internal* states.¹³³

Language use appears to be *neither* deterministic (being stimulus-free) *nor* random (unbounded yet appropriate). How does my novel remark, expressing a novel thought, find coherence in any given situation to your mind if we are both the products of deterministic (or random) internal physiological processes? How do we flexibly use our language in ways that do not veer past the mental states of others, but instead finds convergence with the thoughts of another, sufficient to deem it a ‘meeting of minds,’¹³⁴ without fixed reliance on particular stimuli?¹³⁵

This voluntary action problem is another means by which the competence-performance distinction is derived, dividing the study of language between competence (linguistic knowledge) and performance (linguistic behaviour, which appears not amenable to scientific understanding). It is concomitantly another means by which a *modular* and *generative language faculty*—UG-object—is postulated. Stimulus-free language use indicates that form/meaning pairs are being regulated in the mind/brain without reliance on identifiable stimuli; linguistic resources do not require a pre-specified trigger. Individuals *voluntarily* use language. To account for the cognitive mechanisms enabling this stimulus-free language use, any postulation must meet two conditions: (1) the isolated production of structured linguistic expressions with a discrete set of cognitive resources; and (2) this linguistic production must interface with other cognitive systems (otherwise, expressions would be stimulus-free but effectively random; inappropriate).

A modular and generative language faculty meets both conditions. McGilvray explains:

‘The possibility of stimulus-freedom in language use can be seen to result from a modular language faculty. Not just any kind of modularity will do: we need a faculty that utilises its own resources and with internal prompting produces (in apparent isolation) through its own

¹³¹ Cedric Boeckx, *Language in Cognition* (Wiley-Blackwell 2010) 36.

¹³² Chomsky wrote: “We cannot predict verbal behaviour in terms of the stimuli in the speaker’s environment, since we do not know what the current stimuli are until he responds.” Noam Chomsky, ‘A review of B.F. Skinner’s *Verbal Behavior*’ (1959) 35 *Language* 32.

¹³³ Noam Chomsky, *Cartesian Linguistics* (3rd edn, Cambridge University Press 2009), ch 1.

¹³⁴ John Collins, ‘Between a rock and a hard place’ (2006) 6 *Croatian Journal of Philosophy* 489.

¹³⁵ The problem of internal causation is the same type of problem as external causation. In either case, the expression purportedly being caused (having some identifiable trigger that reliably leads to a certain effect) nonetheless coheres with the mental states of others and is judged to be appropriate to the given situation.

algorithms items in the form of linguistic expressions that are unique to it and yet that “interface” with relevant other internal biological systems.’¹³⁶

Language use is an autonomous operation in that linguistic resources can be recruited without fixed reliance on stimuli, for any purpose or situation, which indicates that these resources are isolated. In other words: modular and domain specific.

This is a fuller description of the internally directed behaviour of human children outlined above in response to CVCL. Indeed, voluntary action raises problems for computational modelling: both the Theory and Proxy View must account for the cognitive mechanisms necessary to establish the possibility of stimulus-free and unbounded yet appropriate linguistic behaviour. They must offer a *better explanation* than the postulation of a modular and generative language faculty.

The LLM Theory position is inadequate, as LLMs are stimulus-controlled; their outputs have identifiable stimuli (internally given custom instructions or externally given inputs in the form of prompts). They are input-output mechanisms. As a result, the LLM Theory view does not offer a *better explanation* for the cognitive architecture supporting stimulus-free language use, as it offers *no explanation* at all.¹³⁷

Proxy Views are also in a bind. The dominant position is that transformer-based and other machine learning models, while weakly biased (in that they are not literal blank slates), are *not* modular and *not* domain-specific. Contreras Kallens et al. endorse the view that LLMs acquire their abilities without language-specific constraints.¹³⁸ Millièrè similarly argues: ‘While...Transformers lack language-specific innate knowledge, they have more domain-general biases.’¹³⁹ He continues in a footnote: ‘Insofar as vanilla language model architectures do not have language-specific inductive biases, their tentative success in realistic learning scenarios may count as evidence against a strongly modular language faculty as postulated by Chomsky.’¹⁴⁰

It is not clear on what basis this claim can be made. The voluntary action problem *limits* the usefulness of Proxy Views, as they cannot lay claim to insights about the cognitive architecture of human language. The stimulus-freedom of language use *does* entail both modularity and domain-specificity. If computational models are non-modular and domain-general, they are poor proxies for theories of human linguistic cognition.¹⁴¹

7. The Upshot on Computational Modelling

Computational modelling risks *extending* disputes of a fundamental nature about language or morality rather than mitigating or resolving them. This is not surprising: if one conceives of

¹³⁶ James McGilvray, ‘Chomsky on the creative aspect of language use and its implications for lexical semantic studies’ in Federica Busa and Pierette Bouillon (eds.), *The Language of Word and Meaning* (Cambridge University Press 2001) 12.

¹³⁷ See, Vincent J. Carchidi, ‘Creative minds like ours?’ (2024) 18 *Biolinguistics* 1-31.

¹³⁸ Kallens et al. (n 87) 2.

¹³⁹ Raphaël Millièrè, ‘Language models as models of language’ [2024] arXiv 32.

¹⁴⁰ *Ibid.*

¹⁴¹ For extended engagement, see, Vincent J. Carchidi, ‘The creative aspect of human language use’ in José-Luis Mendivil-Giró (ed.), *Artificial Knowledge of Language* (Vernon Press, In Press).

language as sociohistorical constructions like English or Swahili, then one will find the modelling of specific languages a fruitful, even exhilarating activity. If language is an object internal to the mind, then computational modelling is more likely to be a somewhat limited proxy.

Machine learning models in particular present sharp disanalogies between themselves and humans that keep this disagreement roughly where it was. These include the *format* of the input, the *sequence* of the learning trajectory, the *different constraints* afforded by the learning objectives, the *means* by which the data over which learning operates is *collected* (i.e., autonomously by the human child or otherwise human-given), and the *nature of the use* of acquired linguistic or other knowledge (stimulus-free, unbounded, and appropriate in humans).

That said, there are means by which the Proxy View can be maintained, having noted these existing disanalogies. Such modelling must be properly *controlled*, such that heavy cognitive lifting is not unwittingly done on the model's behalf. Such a model must deal with the appropriate *format* of human cognition and must be appropriately constrained during training. It must likewise be shown to possess *knowledge* comparable to an individual's in the linguistic or moral domain, attained via comparable *environmental exposure* (e.g., amount and kinds of data) and *developmental trajectory*. Voluntary use, for its part, may be off-limits to computational models indefinitely.¹⁴²

To overcome fundamental barriers posed by matters including autonomous data collection (akin to a human child's self-directed curiosity) and learning without external instructions (i.e., without a training objective provided from the outside) would seem beyond the limits of machine learning models and foreseeable computational models (though, we do not rule it out). This appears to make the use of such models as theories in themselves untenable.

8 The Theory-Dependence of Simulations

This leads us to the importance of what Mahlmann calls the 'theory-dependence of data interpretation' in his critique of the 'mental gizmo thesis' articulated chiefly by Greene.¹⁴³ Mahlmann notes, in the context of neuroimaging, that '[d]ata only have meaning within a theoretical framework...If the framework is deficient, the interpretation of the data will be insufficient, too.'¹⁴⁴

The use of computational models as proxies for existing theories implies that the researcher has already determined the relevant facts of the phenomenon and likewise determined which data—if any particular subset of data is weighed more heavily—is to be given explanatory force. The simulation of cognitive acquisition by machine learning models is an *experiment*. Experiments are theory-dependent endeavours.

Note that this applies even if one rejects the generative approach to language or morality. There is a fundamental divide. Mendiñil-Giró identified the division: usage-based approaches are *inductive*, where theory is derived *from* data, and the more data, the more robust the theory. UG-theory (and

¹⁴² See, Vincent J. Carchidi, 'Rescuing mind from the machines' (2025) 168 *Philosophy Now* 12-16.

¹⁴³ Greene, *Moral Tribes* (n 36).

¹⁴⁴ Mahlmann, *Mind and Rights* (n 1) 346.

UMG-theory, we add) is *deductive*, treating the *discovery* of data and the *justification* of a theory as separate matters, with the theory being a *conjecture* whose underlying means are irrelevant.¹⁴⁵ Yet, computational models, while adequate to serve as proxies for those theories that *could* refute UG or UMG, cannot resolve this divide on their own. More specifically, there are two problems: (1) humans and models exhibit fundamental disanalogies, most prominently revolving around the internally-directed, autonomous acquisition of cognitive abilities by humans; and (2) there is no means by which a computational model could intervene on these divergent approaches to science, as determining whether they are adequate proxies requires that one has *already* taken a position on the inductive or deductive approach to cognitive science.

To be sure, in the real course of science and philosophy, positions get crisscrossed; it is not as though each data point is given equal dedication in even the inductive researcher's efforts. The point nonetheless remains: computational modelling is experimental, and experiments are theory dependent.

This does not mean that the interpretation of data is strictly a matter for moral psychology—the bulk of *M&R* builds an analytical theory of morality that is not explicitly about the mind. However, this only heightens the stakes in breaking down a phenomenon like human rights into its constituent parts without losing sight of the fact that there is ‘a specific mental domain of morality, a qualitatively distinguished element of conscious thought, an introspectively accessible, distinctive, intuited, subjectively experienced aspect of our mental life...’¹⁴⁶

9. A Willingness to Be Puzzled About Human Rights

The common thread in the cases of LLMs and CVCL is a persistent underappreciation of the complexity and distinctiveness of the underlying phenomenon—human language. More than this, there is a tendency to assume the relevant facts about language are *obvious*. They lack what Chomsky might call a ‘willingness to be puzzled’¹⁴⁷ about an otherwise familiar phenomenon. At the centre of the mentalist theory is the idea that humans see the world in highly specific and structured moral terms. Mahlmann's claim regarding human rights is that the human normative reasoning from which they result has a *peculiarly* species-specific structure, enabling individuals to accept obligations that arise in relation to the legitimate claims made by others. The logic of moral intuitions, with sufficient abstraction, generalisation, and universalisation, have given rise, in part, to human rights—such rights ‘bind their addressees and thus create legitimate normative burdens for third parties’ yet they are accepted out of ‘reflective self-rule...the fruit of autonomy.’¹⁴⁸ We should take a moment to appreciate how strange an idea this is in recognising that there appears to be no analog for this normative structure in the rest of the animal world nor is there any obvious reason why it should exist at all.

It is an idea that makes itself no less bizarre in the expression of human rights as those which, as Mahlmann illustrates, ‘mark a space of decency defined by human critical thought committed to

¹⁴⁵ José-Luis Mendivil-Giró, ‘How much data does linguistic theory need’ (2019) 3 *Frontiers in Communications* 1-2. Piantadosi seems to agree, noting that his position is ‘inductive.’ Piantadosi, *Language Models* (n 85) 385.

¹⁴⁶ Mahlmann, *Mind and Rights* (n 1) 405.

¹⁴⁷ Noam Chomsky, *What Kind of Creatures Are We?* (Columbia University Press 2016) ch 1.

¹⁴⁸ Mahlmann, *Mind and Rights* (n 1) 79.

obligatory moral principles when reflecting on the nature of the human condition and what is valuable in a human life'¹⁴⁹ (original emphasis removed). Human morality and the rights to which it has given rise do not merely specify what one must and must not do; rather, they specify the characteristics of valued human affairs. We may add to this Dwyer's observation: 'Human beings do not merely believe that certain actions are permissible or obligatory and others not. They *judge* them to be so...' ¹⁵⁰ A strange thing indeed.

It is likewise not obvious that the underlying complexity *M&R* finds in the institutionalisation of human rights would appear in other areas of law, like international contract law. Ivison argues that there is a peculiar *structure* underlying international contract law, particularly embodied by the United Nations Convention on Contracts for the International Sale of Goods, where reciprocity values 'appear to be principled and universally shared.'¹⁵¹ The Convention, she concludes, 'provides evidence of convergence in reciprocity judgment, which can be accounted for within the UMG model.'¹⁵²

All this is perfectly ordinary. As is the most fundamental problem of voluntary language use, barely achieving a handful of references in the sprawling debates about LLMs in linguistics. Yet, very little of what constitutes language and morality turns out to be obvious. The wisdom in the effort undertaken in *M&R* is to methodically structure an analysis of human rights—a familiar and obvious phenomenon—in ways that serve to undercover the richness of human moral thought and the kinds of minds that produce it.

Computational modelling of cognitive acquisition risks avoiding the need for a willingness to be puzzled about something peculiarly human, like morality, and thus the more rigorous approach to theory-construction that this willingness enables. The end-result so often sought after, particularly in the case of linguistics, is the *actualisation* of a cognitive phenomenon—such as human-like linguistic outputs.¹⁵³ Yet, the generative tradition is oriented toward the internal and domain-specific cognitive mechanism that enables these actualisations—for good reason, I have argued. Researchers who use simulations to intervene in theoretical debates about cognitive acquisition thus risk believing that the phenomenon in question is obvious, and what is needed is a way of simply approximating its actualisation.

The challenges posed today by statistical machine learning are vexing without paying due attention to the sophistication of an analytical theory that identifies those aspects of human cognition which link the researcher to a deeper account of this phenomenon. The domain of human rights offers an entry point for articulating this account. *M&R* is a major contribution to this end—and one which should continue to be hotly debated.

¹⁴⁹ Ibid 280.

¹⁵⁰ Susan Dwyer, 'Mind your morals' (2008) 2 *Biolinguistics* 111.

¹⁵¹ Harriet Ivison, 'Universal moral grammar and international contract law' (2025) 45 *Journal of Theoretical and Philosophical Psychology* 2.

¹⁵² Ibid 17.

¹⁵³ 'We only get to principles by seeing if the simulations recapitulate the same high-level properties of the system we're interested in.' Piantadosi (n 85) 380.